

Chromatography

Learning outcomes

By the end of this section you should be able to:

- Identify the difference between a dye and a pigment.
- Outline the process of chromatography in terms of phases and intermolecular forces acting.
- Explain, calculate and interpret the retardation factor values, R_F .

When making decisions about the food we eat, we use our senses to assess smell, taste and to determine the texture and appearance. One important aspect of appearance is the colour of the food. Knowing this, some food producers enhance the colour of foods using additives, such as synthetic dyes or natural pigments from plants.

In the past, some food colourings have been discovered to be dangerous to health. Some pigments and dyes were even found to be carcinogenic. How can regulatory bodies, such as the World Health Organisation (WHO), be sure that the additives in different foods are safe?

Dyes versus pigments

The terms dye and pigment are often used interchangeably in everyday life as colouring agents for foods and art supplies, but have different meanings in chemistry. Dyes are typically synthetically made whereas pigments are derived naturally from plants (**Figure 1**).

Dye molecules are typically small, polar molecules. On the basis of your knowledge from [section S2.2.9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-intermolecular-forces-id-43852/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-intermolecular-forces-id-43852/), do you think dye molecules are likely to dissolve in water? Compare this to pigment molecules, which are generally much larger, non-polar molecules. Do you think pigment molecules are likely to dissolve?

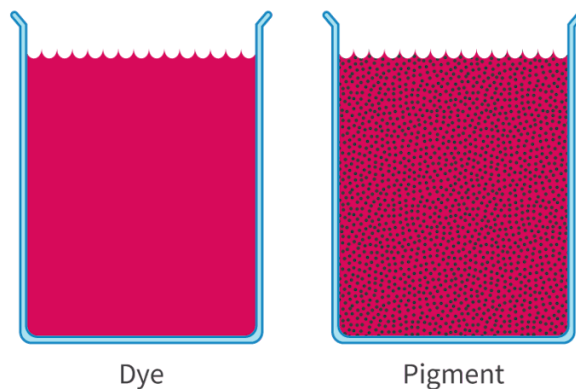


Figure 1. Dyes vs pigments.

 More information for figure 1

The image shows two separate containers. The first container, labeled "Dye," appears to be filled with a uniform, solid color, indicating a transparent solution of dye fully dissolved in the liquid. The second container, labeled "Pigment," is filled with a speckled pattern, representing the suspended particles of pigment in the solution. This illustrates the difference in behavior when dyes and pigments are mixed with a solvent —dyes dissolve to form a transparent solution, while pigments remain suspended, creating an opaque mixture.

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Dye molecules will typically dissolve in a polar solvent such as water and produce a transparent solution, similar to **Figure 1**. Pigment molecules do not do this. When mixed with a solvent the pigment molecules will remain suspended in the solution, producing an opaque solution. To dissolve pigments, often another chemical known as a ‘binder’ is added to the pigment. **Table 1** below provides a summary of the properties of dyes and pigments.

Table 1. Properties of dyes and pigments.

Property	Dye	Pigment
Molecular size	Small	Comparatively large
Solubility	Soluble in water	Insoluble in water and many other solvents
Impact on solution	Transparent	Opaque

Most coloured substances are composed of a range of dyes or pigments. For example, in just plants there are four categories of pigments; chlorophyll, carotenoids, xanthophylls and flavonoids. Some of these categories contain

hundreds of different pigments! **Figure 2** shows an example of the different pigments in just one type of vegetable – carrots.

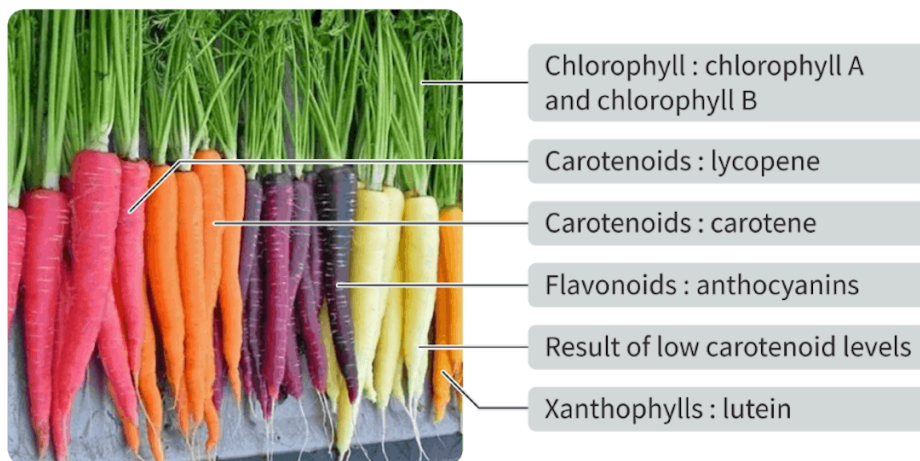


Figure 2. Carrots naturally contain a range of different pigments.

 More information for figure 2

The image shows a variety of carrots in different colors, each labeled to indicate the type of pigment present. The carrots are arranged from left to right with labels pointing to specific ones: "Chlorophyll: chlorophyll A and chlorophyll B" refers to the green parts of carrots. "Carotenoids: lycopene" points to the red carrots. "Carotenoids: carotene" corresponds to the orange carrots. "Flavonoids: anthocyanins" is associated with the purple carrots. A note "Result of low carotenoid levels" describes the lighter colored carrots, and "Xanthophylls: lutein" refers to the yellow carrots. This presentation highlights the diverse pigmentation found in carrots.

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While the pigments in carrots occur naturally and have many health benefits, dye in other synthetic foods do not. Chromatography can be used to identify the different pigments present in foods and other coloured substances.

Chromatography

Section S1.1.1b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/>) introduced chromatography as a technique used to separate the components of a mixture. The components, once separated, can then be identified using other techniques. Students are required to know about two types of chromatography - paper chromatography and thin layer chromatography (TLC). This section will mainly focus on paper chromatography; thin layer chromatography is covered in more detail in section 1.1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/paper-or-thin-layer-chromatography-id-48918/>).

Chromatography is a qualitative analytical process, it is not an example of quantitative analysis as it does not determine how much of each substance is present.

The chromatography technique is composed of two phases: the stationary phase and the mobile phase. The components of the mixture will either have an attraction for the mobile phase or stationary phase, based on the intermolecular forces present, which allows for effective separation. The next section will focus on the role of these two phases and the use of paper chromatography to separate the components of a mixture.

Experimental technique

1. A pencil line is drawn on the piece of chromatography paper.
This acts as a start point so it is clear how far the mixture will have travelled. The piece of chromatography paper is known as the stationary phase as this medium does not move (**Figure 3**).
2. The mixture which is to be separated is placed on the pencil line on the paper. Very small spots are added.
3. An appropriate solvent is selected and the end of the paper is placed in the solvent. The solvent line should not reach the pencil line. The solvent will travel up the paper and carry the mixture with it. Because this medium moves, it is called the mobile phase.
4. Once the mobile phase (the solvent) has travelled up the paper and the mixture has been separated, the paper is removed and left to dry.
5. The chromatogram produced is compared to a set of known values and components are identified.

Chromatography and intermolecular forces

At the beginning of the chromatography process a small amount of the mixture is added to a pencil line at the bottom of the paper. Why do you think the line must be drawn with a pencil and not in pen?

If the line was drawn in ink then it would also dissolve in the solvent and travel up the paper too.

The distance the solvent has travelled from the pencil line to where it stopped moving is called the solvent front.

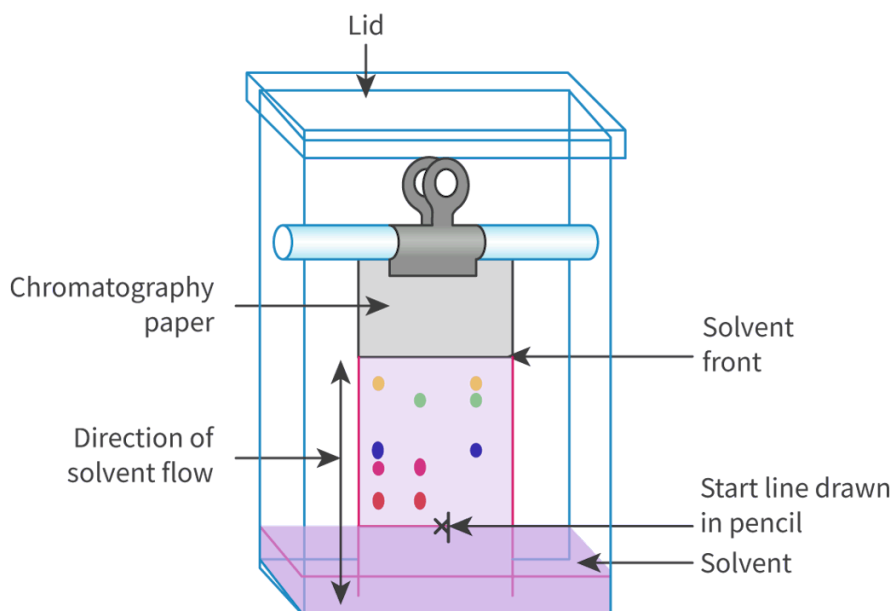


Figure 3. Chromatography apparatus.

 More information for figure 3

The image shows a diagram of a chromatography apparatus. It is a rectangular, transparent container with a lid at the top. Inside, a piece of chromatography paper hangs vertically. The paper is secured at the top by a clip. Various colored dots are distributed along the chromatography paper, indicating different substances traveling up the paper. The bottom of the apparatus contains a solvent in which the bottom edge of the chromatography paper is immersed. Arrows on the diagram indicate the direction of solvent flow upward through the paper. The solvent front is marked near the top portion of the paper, demonstrating how far the solvent has traveled from the starting line drawn in pencil at the bottom of the paper. Labeled parts include 'Lid,' 'Chromatography paper,' 'Solvent front,' 'Start line drawn in pencil,' 'Solvent,' and 'Direction of solvent flow.'

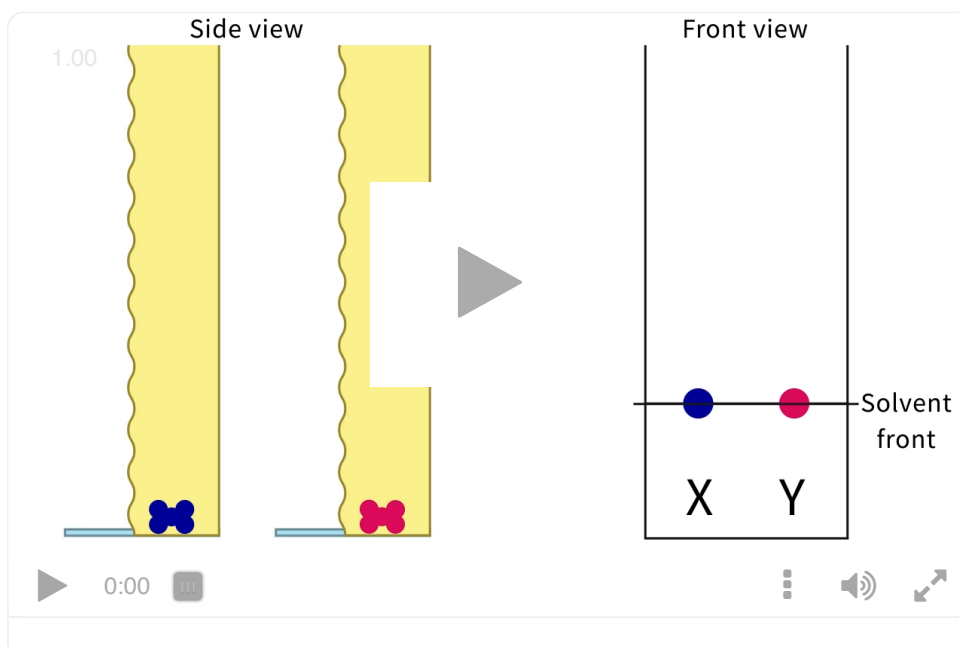
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The stationary phase is a piece of chromatography paper that has a solvent trapped between its fibres. In thin layer chromatography, the stationary phase is usually silica gel, a substance produced by combining sodium silicate and sulfuric acid. The mobile phase is either a polar solvent such as water, or a non-polar solvent such as hexane, depending on which one the mixture will dissolve in. The components of the mixture will either travel with the solvent if they are alike (i.e. both polar or both non-polar) or get trapped in the pores of the paper if they are not alike (i.e. a polar mixture with a non-polar solvent or vice versa).

Table 1. Stationary and mobile phases.

Stationary phase	Mobile phase
Chromatography paper	Solvent that dissolves the mixture being analysed

The mixture will travel up the chromatography paper as it is attracted by the water molecules trapped in the paper's fibres. It uses the **intermolecular forces** that have caused it to become adsorbed on the surface of the paper. It will exist in equilibrium between the hydrogen bonds it forms with the solvent (the mobile phase) and the intermolecular forces it forms with the paper (the stationary phase). This will determine how far up the paper it travels: the amount of time spent in each of the phases is a reflection of the strength of the intermolecular forces that are experienced in each phase (**Interactive 1**).



Interactive 1. Paper chromatography can separate mixtures based on the attraction of its components for the mobile phase and the stationary phase.

 More information for interactive 1

This interactive video demonstrates the concept of paper chromatography, a technique to separate and identify components in a mixture. This interactive begins by showing the setup from both side and front views. A test sample is placed on chromatography paper, which acts as the stationary phase. A solvent, such as water or ethanol (mobile phase), moves upward on the chromatography paper through intermolecular force. In the side view, two chromatography papers are placed. In both papers, the solvent moves upward with two solvents moving upward. The substances are represented as blue dots X and red dots Y. Blue dot substance X in the first chromatography paper moves slowly along with the movement of solvent, and red dot substance Y in the second chromatography paper moves faster along with the movement of solvent (mobile phase). The text appears in the video as the red dot substance moves upwards and reads, the sample moves from the stationary phase to the mobile phase. In the front view, chromatography paper shows both substances, the blue dot substance X and the red dot substance Y. A horizontal line is marked as the solvent front. As the solvent front moves along with the substance, X and Y move forward. Substance X moves slowly, whereas substance Y moves quickly.

This interactive both side view and front view end by displaying the final positions of the substances on the paper, showing how different components can be visually separated and identified. The text appears with an arrow marking the substance Y on the side view and reads, this substance has moved farther because it is more attracted to the mobile phase.

This interactivity helps the user to differentiate between the movement of the stationary phase (paper) and the mobile phase (solvent). This method also explains how intermolecular forces affect the movement of substances on paper.

If a substance has strong intermolecular forces with the stationary phase, then it will remain adsorbed for much longer than if it has weak intermolecular forces. Therefore, compounds with different strengths of intermolecular force will move at different rates.

Study skills

Try not to confuse **adsorbed** with the term **absorbed**. Adsorption occurs when a chemical species adheres to the surface of a material, while absorption occurs when they penetrate the inside of another material (**Figure 4**).

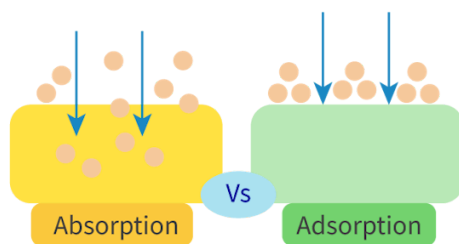


Figure 4. Absorption vs. adsorption.

 More information for figure 4

This is a diagram illustrating the difference between absorption and adsorption. The image is divided into two sections. On the left, labeled "Absorption," a yellow rectangle represents a material that particles are penetrating, symbolizing absorption. Arrows point downward into the material, demonstrating penetration. On the right, labeled "Adsorption," a green rectangle represents a material where particles are adhering to the surface, symbolizing adsorption. Arrows point toward the surface of the rectangle, indicating adhesion. The terms "Absorption" and "Adsorption" are clearly labeled under each respective diagram, and a "Vs" symbol is in the middle, highlighting the comparison between the two processes.

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Interpreting the chromatogram

Paper chromatography produces both quantitative and qualitative data, both of which are important.

Practical skills

Tool 1: Experimental techniques — Measuring variables

Inquiry 2: Collecting and processing data — Processing data

It is possible to underestimate the importance of collecting both quantitative and qualitative experimental data. Although quantitative measurements are more objective than qualitative measurements, both types should always be collected in experiments.

At first glance, the number of spots will indicate the purity of a substance. If only one spot is produced then the substance is pure. If two or more spots are produced, then the substance is impure. This is a good use of qualitative data.

When comparing an unknown mixture against a reference sample, the number, colour and location of the spots can be used. If an unknown has the same number of spots, in the same location and they are the same colour, then it is probably the same substance. This is another good use of qualitative data.

The retardation factor, also known as retention factor, or R_F value can also be calculated. This value represents the distance the spot has travelled relative to the distance the solvent has travelled, and can be compared against other known values. This value is useful as it allows you to identify spots you may not have tested at that time so a direct comparison with a reference is not possible. The R_F value for a substance will always be the same regardless of how much time the solvent was given to travel, once the same mobile and stationary phases are used.

The R_F value is calculated using the formula:

$$R_F = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

R_F values range from 0 to 1. Why are different numbers produced?

A value close to 0 occurs when the substance has very low attraction to the mobile phase and high attraction to the stationary phase. The closer the value gets to 1, the higher the attraction of the substance to the mobile phase and the lower the attraction to the stationary phase. Calculation of retardation factor supports the accurate identification of substances, but does not quantify how much of that substance exists. Again, demonstrating that paper chromatography involves qualitative data not quantitative data.

Worked example 1

A sample of food dye used to colour different candies was analysed using paper chromatography.

Calculate the R_F value for each of the food colourants extracted from the food dye.

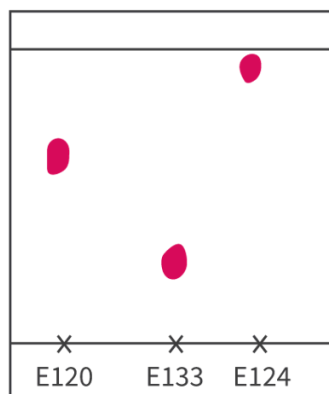
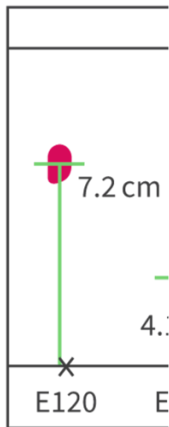


Figure 5. A sample of food dye was analysed.

 More information for figure 5

The image is a chromatogram depicting the analysis of a sample of food dye. It features a rectangular box with three labeled x-axis markers: E120, E133, and E124. Above each of these markers, there are distinct, circular spots in red, representing the food dye's movement on the chromatogram. E120 is on the left, E133 in the middle, and E124 on the right. This setup indicates that the dyes have been separated based on their differing properties, and each spot's position implies a different retention or migration factor (R_F value), which is typically calculated for each dye during analysis. The spots are placed at various heights above their labels, suggesting their relative mobility in the chromatographic process.

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Solution steps	Calculations
Step 1: Measure the distance each spot has travelled and the solvent front.	 The diagram shows a vertical rectangular plate representing a TLC plate. A horizontal line at the bottom is labeled 'E120' and 'E'. A red spot is located at a distance of 7.2 cm from the baseline. A green line extends from the baseline to the spot, with a label '7.2 cm' next to it. Another green line extends from the baseline to a horizontal line representing the solvent front, with a label '4.1' next to it.
Step 2: Calculate the R_F value for each using the formula:	$R_F = \frac{\text{distance travelled by sample}}{\text{distance travelled by solvent front}}$

Substance	Distance moved by sample (cm)	Distance moved by solvent front (cm)
E120	7.2	10.0
E133	4.1	10.0
E124	9.5	10.0

Practical skills

Tools 1: Experimental techniques — Applying techniques

There are many other chromatography methods in addition to paper chromatography. The choice of method depends on the properties of the mixture.

- Thin layer chromatography (TLC) is similar to paper chromatography, except that it uses a thin glass or plastic plate containing a layer of silica or alumina as the stationary phase. You can find more information about thin layer chromatography in section [1.1.4](#)

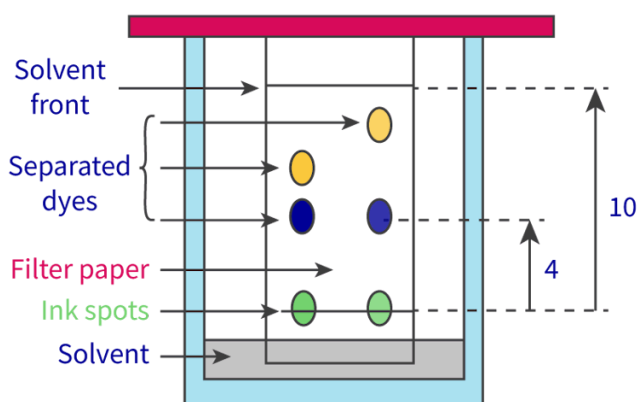
(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/paper-or-thin-layer-chromatography-id-48918/>).

- Gas chromatography (GC) uses an inert gas as the mobile phase to separate highly volatile mixtures.
- High performance liquid chromatography (HPLC) uses a high-pressure pump to force a liquid mobile phase through a stationary phase consisting of a column of small beads. This allows for a faster separation of the mixture.

Summary

Paper chromatography is a technique used to separate and identify components in a mixture. The mixture is dissolved in a solvent that travels up the paper, known as the mobile phase. The chromatography paper is the stationary phase (**Figure 6**).

The relative affinity of the components of the mixture, based on the strength of intermolecular force formed, for each of these phases is what determines the rate at which they travel up. The paper containing these separated components of the mixture is known as a chromatogram. R_F values for each component are calculated and compared to a sample of known values.



 More information

The diagram illustrates the process of chromatography. It features a setup with a container filled with a solvent at the bottom. A filter paper strip extends vertically from the solvent into the air. The diagram labels several components: the solvent at the bottom, the filter paper extending upward, and the solvent front at the top of the wet portion of the paper. Dyes are separated along the paper, with colored spots representing separated components moving up from the bottom. The sequence of colors includes yellow, blue, and green ink spots. The text indicates the heights at which the different elements are positioned: the solvent front at the top, separated dyes in the middle, and solvent at the bottom. There are measurements on the right side, indicating a total height of 10 units and a segment of 4 units from the baseline up to the ink spots, illustrating the movement of the dyes relative to the total solvent travel.

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Credit: Sinhyu, Getty Images

Figure 6. Paper chromatography.

In this activity you will have the opportunity to apply the knowledge you have learned about soluble mixtures by analysing a chromatogram obtained by your teacher. Remember to use key terminology you have learned throughout this section in your answer.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Asking questions and framing hypotheses based upon sensible scientific rationale
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Case study

While you were completing the most recent Chemistry test, the teacher had to leave the room for a moment. During this time, you saw two other students swap exam papers and continue writing on each other's tests. Your teacher is suspicious of the grades one of these students achieved and is offering a reward if any student can prove that cheating occurred. The teacher suspects that three

other students may have been involved in the incident and collected the pens used by all students to write the tests. Answer the questions to show whether cheating happened and to make sure the teacher knows you were not involved!

Figure 7 shows the chromatogram the teacher obtained when testing the students' pens.

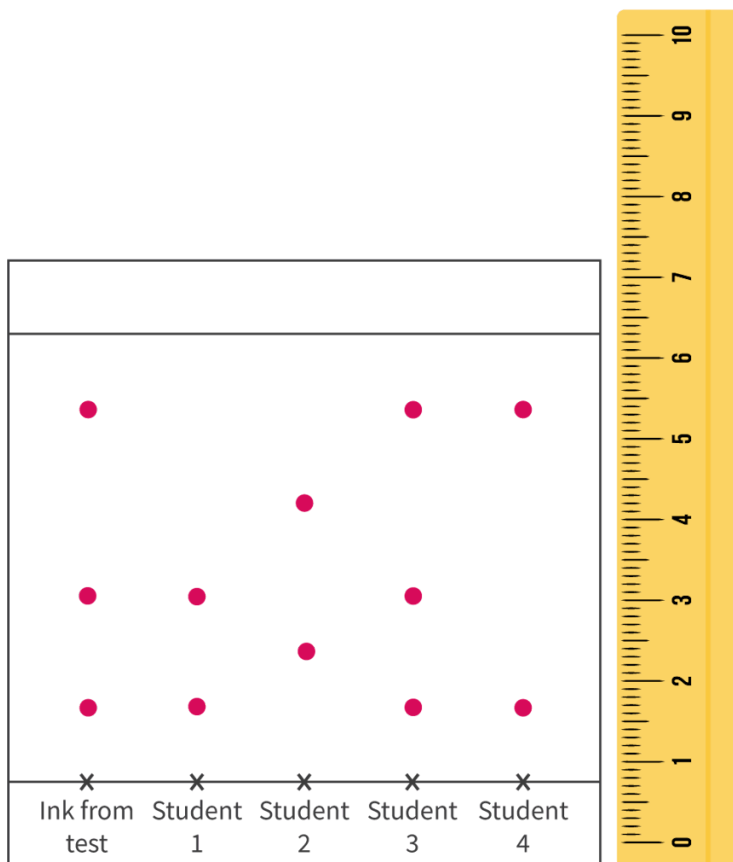


Figure 7. The teacher's chromatogram.

 More information for figure 7

The image shows a chromatogram used to test ink samples from different sources, including an ink referred to as "Ink from test" and samples from four students (1 to 4). Each ink source is plotted along the X-axis. The chromatogram features several pink dots, representing the ink's migration at various distances. A ruler on the right side of the image provides a scale for measuring the ink spots. The height of the dots indicates the migration distance, facilitating analysis of the ink samples' components.

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Questions:

1. Describe how the chromatogram could be attained by the teacher.
2. Identify how many different dyes were in the ink from the test?
3. Identify which student's pen was definitely not used to cheat on the test. Justify your choice.
4. Suggest which student was involved in the cheating. Justify your choice.
5. When doing chromatography, it is important that the line on which the pens are all placed at the beginning is drawn in pencil and not pen.

Explain why this is using your knowledge of intermolecular forces that was presented in this section.

6. The inks from the suspects pens were compared to a table to known pigments commonly present in pens, shown in **Table 2**. Identify which inks were present in the pen used to write the test. Outline why comparing the values obtained by the teacher against the values obtained from a different chromatogram is valid.

Table 2. R_F values for known pigments.

Ink colour	
Black	
Blue	
Green	
Red	
Yellow	